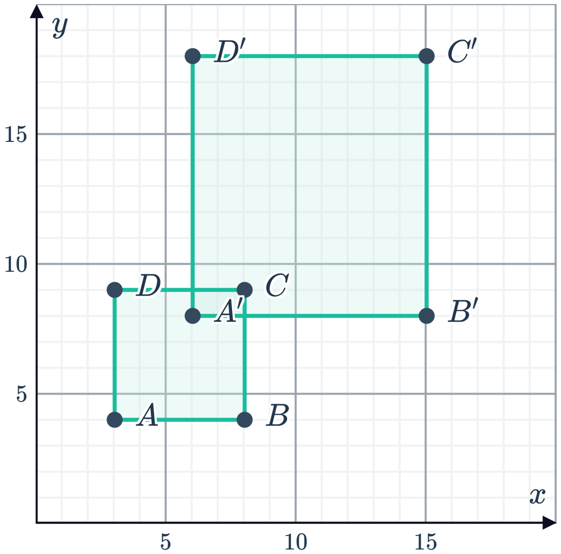
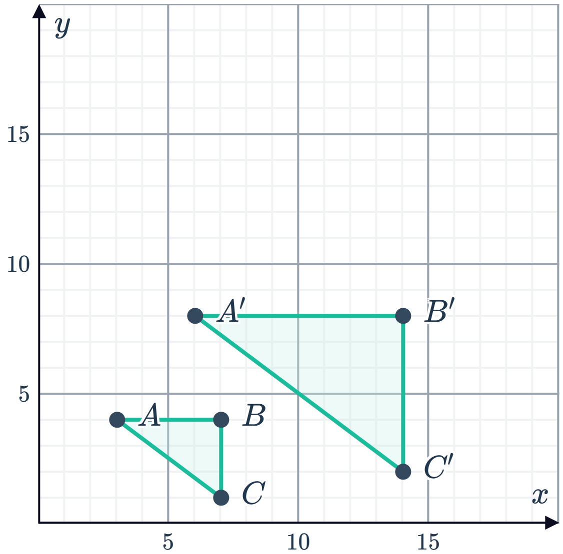
Dilations

1 Determine whether the following graphs show a dilation. If yes, find the scale factor.

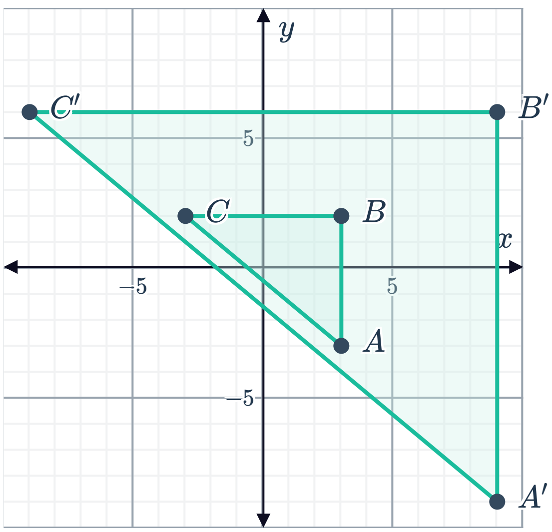
a



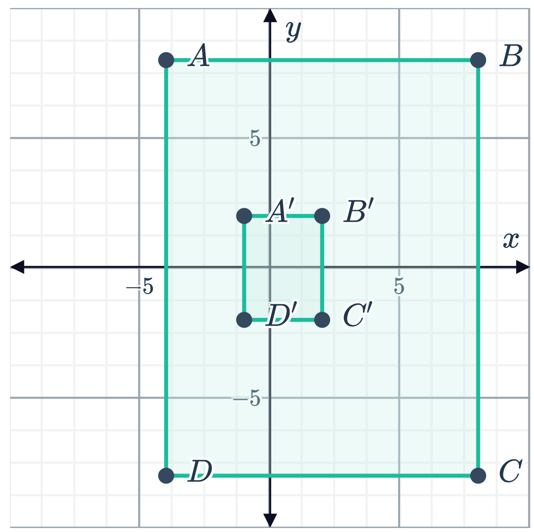
b



c

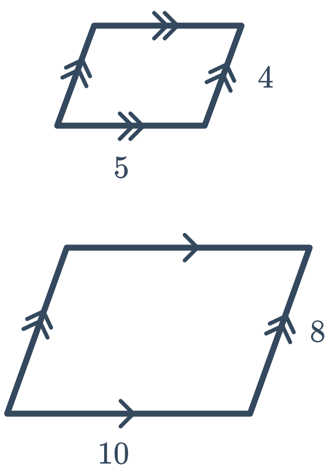


d

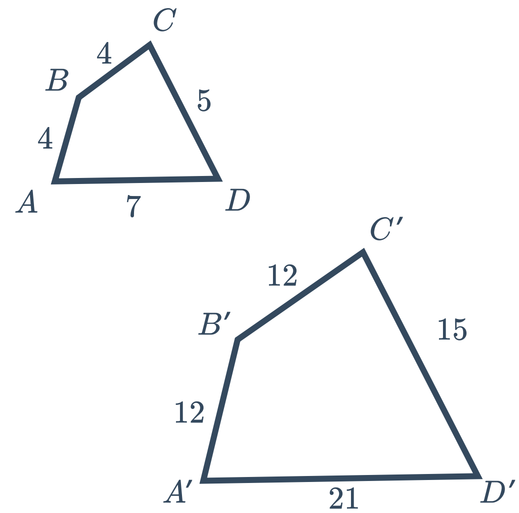


2 Find the scale factor for each pair of similar [redacted]es:

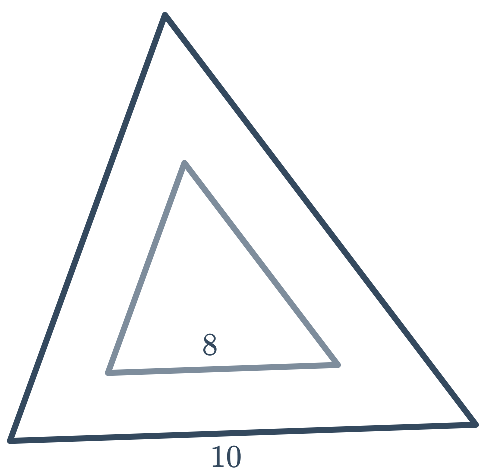
a



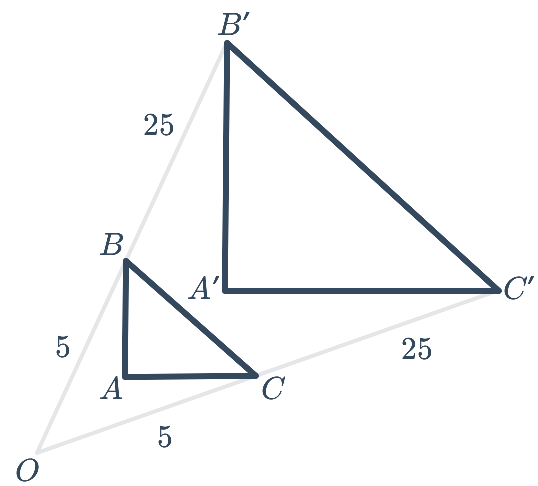
b



c

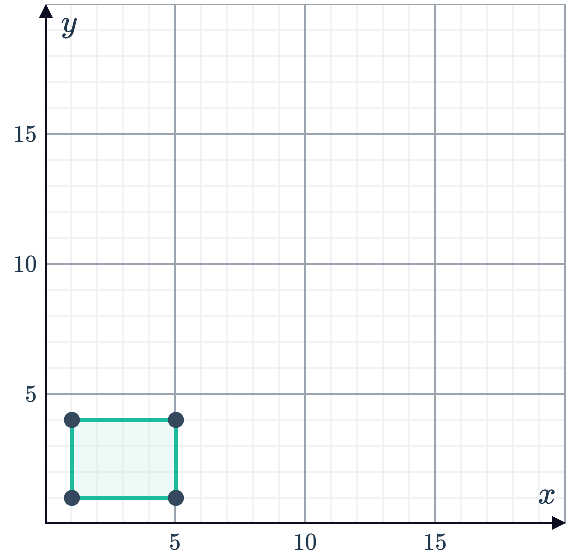


d

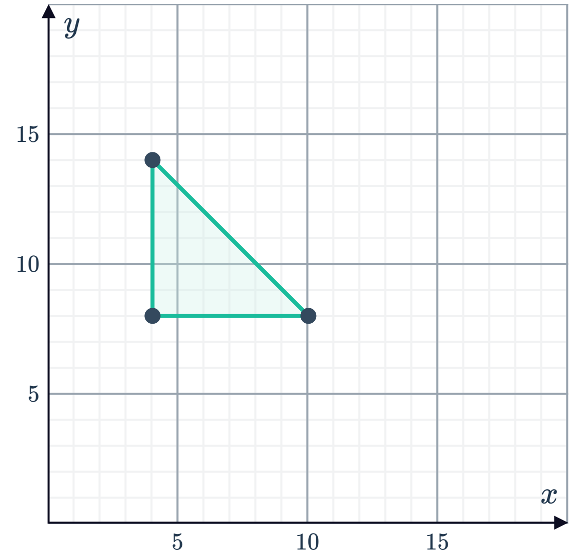


3 Dilate the figure by the given factor using the origin as the center of dilation:

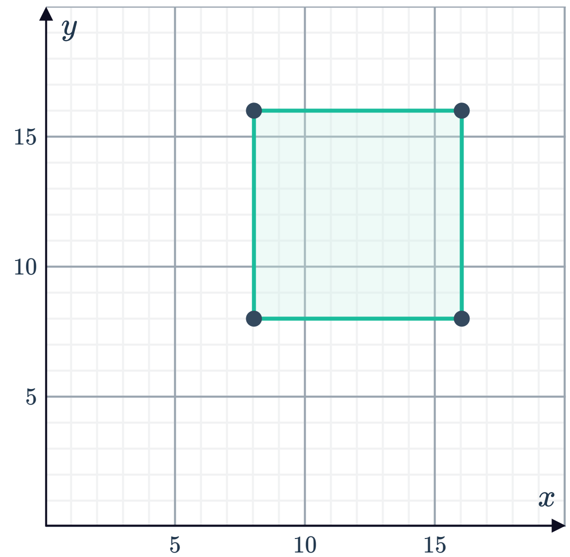
a Scale factor of 3



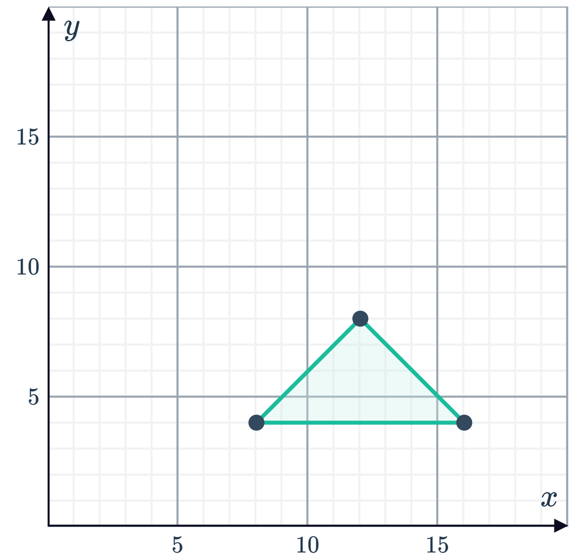
b Scale factor of 4



c Scale factor of $\frac{1}{2}$



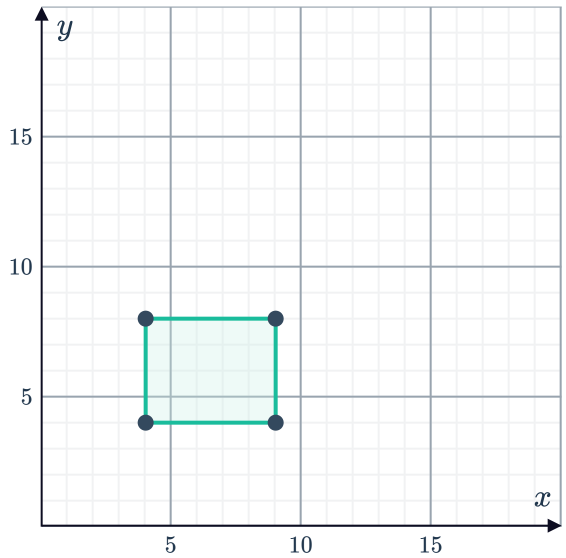
d Scale factor of $\frac{5}{4}$



4 Dilate each figure using the given scale fact[redacted]d center of dilation.

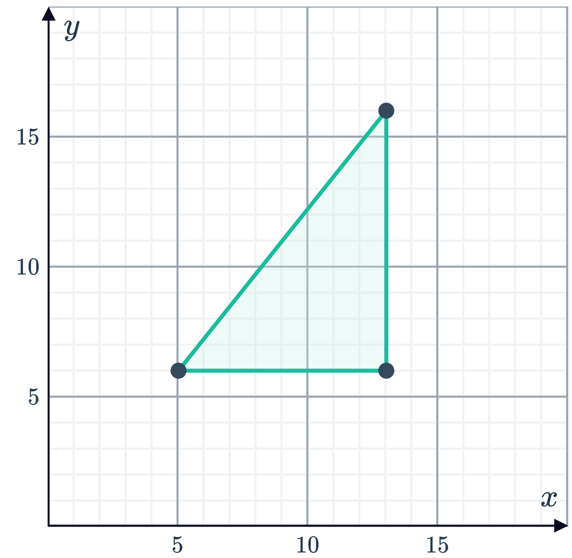
a Scale factor: 2

Center of dilation: $A\,(2, 3)$



b Scale factor: $\frac{1}{2}$

Center of dilation: $A\,(3, 2)$



5 A triangle with vertices $A\,(-7, -5)$, $B\,(3, -5)$ and $C\,(-5, 9)$ is dilated by a factor of 0.5 using the origin as the center of dilation. What are the coordinates of the vertices of the dilated triangle?

6 A rectangle with vertices $P\,(-5, 3)$, $Q\,(3, 3)$, $R\,(3, -5)$ and $R\,(-5, -5)$ is dilated using the origin as the center of dilation. The vertices of the new rectangle are $P'\,(-25, 15)$, $Q'\,(15, 15)$, $R'\,(15, -25)$ and $S'\,(-25, -25)$.

Find the scale factor.